

Edge-antipodal colorings of cubes

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L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

We let Q_d denote the d -dimensional cube graph. A map $\phi : E(Q_d) \rightarrow \{0, 1\}$ is called edge-antipodal if $\phi(e) \neq \phi(e')$ whenever e, e' are antipodal edges.

Conjecture If $d \geq 2$ and $\phi : E(Q_d) \rightarrow \{0, 1\}$ is edge-antipodal, then there exist a pair of antipodal vertices $v, v' \in V(Q_d)$ which are joined by a monochromatic path.